PADHAKU
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CHAPTER 2

Lines and Angles

NCERT Exercise Solutions

In-text questions · 2.2 Line Segment and 2.3 Line	2 Q
Figure it Out · 2.4 Ray	6 Q
In-text questions · 2.5 Angle	4 Q
Figure it Out · 2.5 Angle	6 Q
In-text questions · 2.6 Comparing Angles	5 Q
Figure it Out · 2.6 Comparing Angles	4 Q
In-text questions · 2.7 Making Rotating Arms	3 Q
In-text questions · 2.8 Special Types of Angles	3 Q
Figure it Out · 2.8 Right Angles	4 Q
In-text question · Classifying Angles	1 Q
Figure it Out · Classifying Angles	4 Q
In-text questions · 2.9 Measuring Angles	7 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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In-text questions · 2.2 Line Segment and 2.3 Line

? **Mark two points A and B. What is the shortest route from A to B?**

✔ ANSWER

The straight path from A to B is the shortest route.

This straight path (with A and B) is called the **line segment AB**.

? **Do you think you can draw a complete picture of a line? Why?**

✔ ANSWER

No. A line goes on forever in both directions, without any end.

We can only draw a part of it on paper.

Figure it Out · 2.4 Ray

1 **Rihan marked a point on a piece of paper. How many lines can he draw through the point? Sheetal marked two points. How many different lines can she draw through both points?**

✔ ANSWER

Rihan: **many (countless) lines** can pass through one point.

Sheetal: **only one line** can pass through two given points.

2 **Name the line segments in Fig. 2.4 (points L, M, P, Q, R). Which of the five marked points are on exactly one of the line segments? Which are on two?**

✔ ANSWER

Line segments: **LM, MP, PQ, QR**.

On exactly one segment: **L and R** (end points).

On two segments: **M, P and Q**.

3 **Name the rays shown in Fig. 2.5. Is T the starting point of each of these rays?**

✔ ANSWER

Rays: **TA, TB, TN and NB**.

No. T is the starting point of TA, TB and TN, but ray NB starts at N.

4 **Draw a rough figure and label: a. OP and OQ meet at O. b. XY and PQ intersect at M. c. Line l contains points E and F but not point D. d. Point P lies on AB.**

💡 SAMPLE ANSWER

a. Draw two rays OP and OQ starting from the same point O.

b. Draw two lines XY and PQ crossing each other; mark the crossing point M.

c. Draw a line l; mark E and F on it and D above (not on) the line.

d. Draw line segment AB and mark point P somewhere on it.

5 **In Fig. 2.6, name: a. Five points b. A line c. Four rays d. Five line segments**

✔ **ANSWER**

a. **D, E, O, B, C**

b. **Line DB** (also called DE, DO, EO, EB or OB)

c. **OC, OB, OE, OD**

d. **DE, DO, DB, EO, EB** (OB and OC are also line segments)

6 **Ray OA starts at O and passes through A and B. a. Can you also name it as OB? Why? b. Can we write OA as AO? Why or why not?**

✔ **ANSWER**

a. **Yes.** It starts at O and B is a point on it, so it can be named OB.

b. **No.** AO would be a ray starting at A going towards O. A ray is named starting point first.

❓ In-text questions · 2.5 Angle

? **Vidya opens her book cover (Cases 1–6). Do you see angles in each case? Mark the arms and vertex.**

💡 SAMPLE ANSWER

Yes. The vertex is the spine (fold) of the book.

One arm is the cover, the other arm is the rest of the book (the pages).

? **Which angle is greater — the angle in Case 1 or the angle in Case 2?**

✔ ANSWER

Case 2, because the cover is rotated (turned) more.

? **Compass/divider and scissors: identify the arms and vertex of the angle.**

✔ ANSWER

Compass: arms = the two legs; vertex = the hinge where they are joined.

Scissors: arms = the two blades; vertex = the screw joining the blades.

? **Look at spectacles, wallet and other objects. Identify the angles by marking arms and vertices.**

> 💡 **SAMPLE ANSWER**
>
> Spectacles: the side arm and the frame make an angle at the hinge (vertex).
>
> Wallet: the two flaps make an angle at the fold (vertex).

🧮 Figure it Out · 2.5 Angle

1 **Can you find the angles in the given pictures? Draw the rays forming any one angle and name the vertex.**

> 💡 **SAMPLE ANSWER**
>
> Yes. One angle is **∠BDC**; vertex **D**, arms **DB and DC**.
>
> Find more angles in the other pictures in the same way. (check the picture in your book)

2 **Draw and label an angle with arms ST and SR.**

> 💡 **SAMPLE ANSWER**
>
> Mark point S. Draw ray ST and ray SR from S.
>
> The angle is **∠TSR** (or ∠RST); S is the vertex.

3 **Explain why ∠APB cannot be labelled as ∠P.**

✔ ANSWER

At P there is more than one angle (∠APB, ∠BPC, ∠APC).

So ∠P does not tell which angle we mean. We must use three letters, e.g. **∠APB**.

4 **Name the angles marked in the given figure (points P, Q, R, T).**

💡 SAMPLE ANSWER

∠RTQ and ∠RTP (check the picture in your book)

5 **Mark three points A, B, C not on one line. Draw all possible lines through pairs. How many lines? Name them. How many angles can you name using A, B, C?**

✔ ANSWER

Lines: **3** — AB, BC, CA.

Angles: **3** — ∠ABC (∠CBA), ∠BCA (∠ACB), ∠CAB (∠BAC).

6 **Mark four points A, B, C, D, no three on one line. Draw all lines through pairs. How many lines? Name them. How many angles can you name?**

✔ ANSWER

Lines: **6** — AB, AC, AD, BC, BD, CD.

At each point 3 lines meet, giving 3 angles; 4 × 3 = **12 angles**:

∠BAC, ∠CAD, ∠BAD, ∠ABC, ∠CBD, ∠ABD, ∠BCA, ∠ACD, ∠BCD, ∠ADB, ∠BDC, ∠ADC

❓ In-text questions · 2.6 Comparing Angles

? **Animals opening their mouths: mark the arms and vertex. Arrange the angles from smallest to largest.**

💡 SAMPLE ANSWER

Vertex: the corner where the two jaws meet. Arms: upper jaw and lower jaw.

The wider the mouth is open, the larger the angle. Arrange by how wide each mouth is open. (check the picture in your book)

? **Is it always easy to compare two angles?**

✔ ANSWER

No. When two angles are almost the same size (like 89° and 91°), we cannot compare them just by looking.

We need superimposition or measurement.

? **Here are some angles. Label each angle. How will you compare them?**

💡 **SAMPLE ANSWER**

Label each with three letters, vertex in the middle.

Compare by tracing one angle and placing it over the other (superimposition).

? **Equal angles: ∠AOB and ∠XOY — which is greater?**

✔ **ANSWER**

Neither. On superimposing, OA falls on OX and OB on OY, so the angles are **equal**.

? **Where else do we use superimposition to compare?**

💡 **SAMPLE ANSWER**

To compare lengths of line segments, sizes of squares, circles, coins, leaves, etc.

Figure it Out · 2.6 Comparing Angles

1 **Fold a rectangular sheet, draw a line along the fold. Name and compare the angles between the fold and the sides. Make**

different angles by folding. Which is the largest and smallest angle you made?

💡 SAMPLE ANSWER

Suppose the fold EF goes from side AB to side DC. Angles formed: ∠AEF, ∠BEF, ∠DFE, ∠CFE.

If the fold is slanting, ∠AEF and ∠CFE are larger; ∠BEF and ∠DFE are smaller.

Answers depend on your folds.

2 **Which angle is greater and why? a. ∠AOB or ∠XOY b. ∠AOB or ∠XOB c. ∠XOB or ∠XOC**

💡 SAMPLE ANSWER

a. **∠AOB**, because ∠XOY is only a part of ∠AOB.

b. **∠AOB**, because ∠XOB is a part of ∠AOB.

c. **Both are equal**: OB and OC lie along the same ray, so ∠XOB = ∠XOC. (check the picture in your book)

3 **Which angle is greater: ∠XOY or ∠AOB? Give reasons.**

✔ ANSWER

The two angles look almost the same, so we **cannot tell just by looking**.

We must superimpose (trace and place one on the other) or measure them.

? **Cranes: place the transparent circle on the second crane's angle (OA on OX). Can you now tell which angle is bigger? Which crane was making the bigger angle?**

💡 SAMPLE ANSWER

Point Y lies beyond B on the circle, so the second crane's angle ∠XOY is bigger than ∠AOB.

The second crane opens its mouth wider. (check the picture in your book)

❓ In-text questions · 2.7 Making Rotating Arms

? **Make several rotating arms with different angles and arrange them from smallest to largest using superimposition.**

💡 SAMPLE ANSWER

Place two rotating arms with vertices together and one arm on top of the other.

The one whose second arm turns further is the larger angle. Repeat to put all in order.

? **Which rotating arm will pass through the slit?**

✔ ANSWER

Only the rotating arm whose angle is **equal** to the slit's angle.

The lengths of the arms do not matter.

Challenge **Reduce this angle (by cutting the arms shorter).**

✔ ANSWER

Cutting the arms shorter does **not** reduce the angle.

The size of an angle depends only on the turn between the arms, not on arm length.

? In-text questions · 2.8 Special Types of Angles

? **Straight angle ∠AOB: is it possible to draw OC so that the two angles ∠AOC and ∠COB are equal?**

✔ ANSWER

Yes. Fold the paper so that OB lies exactly on OA. The crease OC divides ∠AOB into two equal angles.

? **Justify why the two angles are equal. Is there a way to superimpose and check? Can it be done by folding?**

✔ ANSWER

When folded, ∠COB lies exactly on top of ∠COA: OB falls on OA and the vertex O stays fixed.

So folding is itself a superimposition, and the two angles are **equal**.

? **If a straight angle is half of a full turn, how much of a full turn will form a right angle?**

✔ ANSWER

Right angle = ½ of a straight angle = ½ × ½ turn = **¼ of a full turn** (a quarter turn).

Figure it Out · 2.8 Right Angles

1 **How many right angles do the windows of your classroom contain? Do you see other right angles in your classroom?**

💡 SAMPLE ANSWER

A rectangular window (or each rectangular pane) has **4 right angles** at its corners.

Other right angles: corners of the door, blackboard, desk top, book, walls and floor.

2 **Join A to other grid points by a straight line to get a straight angle. What are all the different ways?**

💡 SAMPLE ANSWER

Draw the line from A in the direction exactly opposite to AB (continue BA beyond A).

Any grid point on that extended line gives a straight angle. (check the picture in your book)

3 **Join A to other grid points to get a right angle. What are all the ways? (Hint: extend BA to C and divide the straight angle CAB into two equal parts.)**

💡 SAMPLE ANSWER

Join A to a grid point so that the new line makes an 'L' with AB.

On a square grid: if B is 2 right of A, go straight up or down from A; if B is 1 right and 1 down, go 1 right and 1 up (or 1 left and 1 down).

There are two directions (one on each side of AB); every grid point along them works. (check the picture in your book)

4 **Get a slanting crease on the paper. Now get another crease perpendicular to it. a. How many right angles do you have? Justify. b. Describe how you folded the paper.**

💡 SAMPLE ANSWER

a. **4 right angles.** The second fold puts one half of the first crease exactly on the other half, so the two angles on each side are equal; each is half of a straight angle = 90°.

b. Fold the paper once to make a slanting crease. Keep it folded. Now fold again so that the crease lies exactly on itself. Unfold: the two creases are perpendicular.

❓ In-text question · Classifying Angles

? Angles are in three groups; right angles are in the second group. What could be the common feature of the other two groups?

✔ ANSWER

First group: all angles are **less than a right angle** (acute).

Third group: all angles are **more than a right angle but less than a straight angle** (obtuse).

🧮 Figure it Out · Classifying Angles

1 Identify acute, right, obtuse and straight angles in the previous figures.

💡 SAMPLE ANSWER

Less than an 'L' corner → acute; exactly an 'L' → right; more than 'L' but less than a straight line → obtuse; arms in one straight line → straight angle.

E.g. opened book cover on a table (Case of half turn) is a straight angle; scissors slightly open make an acute angle.

2 Make a few acute angles and a few obtuse angles. Draw them in different orientations.

💡 SAMPLE ANSWER

Acute: e.g. 30°, 45°, 70° — draw with the base arm horizontal, vertical or slanting.

Obtuse: e.g. 100°, 120°, 150° — also draw them turned in different directions.

3 **Acute means sharp and obtuse means blunt. Why do you think these words were chosen?**

✔ ANSWER

An acute angle has a narrow opening, so it looks **sharp** like the tip of a needle.

An obtuse angle opens wide, so its corner looks **blunt**.

4 **Find the number of acute angles in each figure. What will be the next figure and how many acute angles will it have? Do you notice a pattern?**

💡 SAMPLE ANSWER

Figures: **3, 12, 21** acute angles.

Each new figure adds 9 acute angles.

Next figure: 21 + 9 = **30 acute angles**. (check the picture in your book)

❓ In-text questions · 2.9 Measuring Angles

? **The circle is divided into 1, 2, 3, 4, 5, 6, 8, 9, 10 and 12 equal parts. What are the degree measures of the angles?**

✔ ANSWER

Parts	Each angle
1	360 ÷ 1 = 360°
2	360 ÷ 2 = 180°
3	360 ÷ 3 = 120°
4	360 ÷ 4 = 90°
5	360 ÷ 5 = 72°
6	360 ÷ 6 = 60°
8	360 ÷ 8 = 45°
9	360 ÷ 9 = 40°
10	360 ÷ 10 = 36°
12	360 ÷ 12 = 30°

? **Unlabelled protractor: using the medium and large marks, can we count the units in 5s or 10s?**

✔ ANSWER

Yes. Long marks come every 10° and medium marks every 5°, so we can count in 10s and 5s.

? Labelled protractor: why does it have two sets of numbers?

✔ ANSWER

So that we can measure from either side.

If the base arm is on the right, read the scale starting with 0 on the right; if it is on the left, read the other scale.

? Name the different angles in the figure (rays OP, OQ, OR, OS, OT, OU) and write their measures.

💡 SAMPLE ANSWER

∠POQ = 35°, ∠POR = 95°, ∠POS = 125°, ∠POT = 160°

∠QOR = 60°, ∠QOS = 90°, ∠QOT = 125°, ∠QOU = 145°

∠ROS = 30°, ∠ROT = 65°, ∠ROU = 85°

∠SOT = 35°, ∠SOU = 55°, ∠TOU = 20° (check the picture in your book)

? Did you include angles such as ∠TOQ? Which set of markings did you use — inner or outer? What is the measure of ∠TOS?

✔ ANSWER

Yes, ∠TOQ = 125°.

Use **one** scale for both arms (the outer scale here).

OT is at 20 and OS at 55 on the outer scale, so ∠TOS = 55 − 20 = **35°**.

? **OT and OS pass through 20 and 55 on the outer scale. How many units of 1° are between them? Can subtraction be used?**

✔ ANSWER

55 − 20 = **35 units**, so the angle is 35°.

Yes, subtraction can be used.

? **Place the protractor with its centre on O and one arm on 0°. What is the degree measure of ∠AOB?**

💡 SAMPLE ANSWER

Arm OB is on the 0 of the inner scale; read OA on the same (inner) scale.

∠AOB = 80° (about) (check the picture in your book)

⭐ Make your own Protractor

2 **The measure of half a turn = ½ of ____ = 180°.**

✔ ANSWER

½ of **360°** = 180°

3 The measure of a ¼ turn = ¼ of 360° = ____. Or, ¼ turn = ½ of 180° = ____.

✔ ANSWER

¼ of 360° = **90°**

½ of 180° = **90°**

4 When folded again, this is ⅛ of a turn, or ⅛ of 360°, or ¼ of 180°, or ½ of 90° = ____.

✔ ANSWER

360 ÷ 8 = **45°**

(also 180 ÷ 4 = 45°, 90 ÷ 2 = 45°)

The other crease gives 180° − 45° = 135°.

5 Continuing with another half fold, we get an angle of measure ____.

✔ ANSWER

½ of 45° = **22.5°**

Think! In Fig. 2.19, ∠AOB = ∠BOC = ∠COD = ∠DOE = ∠EOF = ∠FOG = ∠GOH = ∠HOI = ____. Why?

✔ ANSWER

The straight angle 180° is divided into 8 equal parts by folding in halves three times.

180° ÷ 8 = **22.5°** each.

? **Identify the angle bisectors in your handmade protractor.**

✔ ANSWER

The 90° crease bisects the straight angle (180°).

The 45° and 135° creases bisect the two right angles.

The 22.5°, 67.5°, 112.5°, 157.5° creases bisect the 45° angles.

Figure it Out · 2.9 Measuring Angles (page 35)

1 **Write the measures of the following angles: a. ∠KAL b. ∠WAL c. ∠TAK**

💡 SAMPLE ANSWER

a. **∠KAL = 30°** (yes, we can count in 5s and 10s using the marks)

b. **∠WAL = 50°**

c. **∠TAK = 120°** (check the picture in your book)

Figure it Out · Angle Bisector (page 40)

1 **Find the degree measures of the following angles using your protractor.**

💡 SAMPLE ANSWER

First angle: about **47°**

Second angle: about **23°**

Third angle: about **108°** (check the picture in your book)

2 **Find the degree measures of different angles in your classroom using your protractor.**

💡 **SAMPLE ANSWER**

Corners of the door, board, desk and window: 90°.

Measure others (open door, open book, clock hands) and write them down.

3 **Find the degree measures for the angles given below. Check if your paper protractor can be used here!**

💡 **SAMPLE ANSWER**

First angle: about **42°**; second angle: about **116°** (check the picture in your book)

The paper protractor shows only multiples of 22.5°, so it **cannot measure** these angles exactly; use a standard protractor.

4 **How can you find the degree measure of the angle given below (a reflex angle) using a protractor?**

💡 **SAMPLE ANSWER**

Measure the smaller (unmarked) angle, e.g. 100°.

Marked angle = 360° − 100° = **260°**.

(Or: 180° + the part beyond the straight angle.)
(check the picture in your book)

5 **Measure and write the degree measures for each of the angles a–f.**

💡 **SAMPLE ANSWER**

a. 80° b. 120° c. 60° d. 130° e. 130° f. 60°
(check the picture in your book)

6 **Find the degree measures of ∠BXE, ∠CXE, ∠AXB and ∠BXC.**

💡 **SAMPLE ANSWER**

∠BXE = **115°**

∠CXE = **85°**

∠AXB = 180° − 115° = **65°**

∠BXC = 115° − 85° = **30°** (check the picture in your book)

7 **Find the degree measures of ∠PQR, ∠PQS and ∠PQT.**

💡 **SAMPLE ANSWER**

∠PQR = **45°**, ∠PQS = **100°**, ∠PQT = **150°**
(check the picture in your book)

8 **Make the paper craft as per the instructions. Unfold, draw lines on the creases and measure the angles formed.**

💡 SAMPLE ANSWER

Measure each angle between creases with a protractor.

You will find angles like 90°, 45° and 22.5° made by halving folds.

9 **Measure all three angles of the triangles in Fig. 2.21 (a), (b), (c) and add them. What do you get? Make a conjecture.**

✔ ANSWER

For every triangle the sum comes to **180°** (small errors of 1–2° are from measuring).

Conjecture: **the three angles of any triangle add up to 180°.**

⭐ Mind the Mistake, Mend the Mistake!

? **A student measured ∠U = 35°, ∠V = 80°, ∠W = 70°, ∠X = 150°, ∠Y = 120°, ∠Z = 85°. Identify the incorrect usage of the protractor in each and how to correct it.**

💡 SAMPLE ANSWER

Check each figure for these mistakes: (check the picture in your book)

- **Wrong scale read:** one arm is on 0 of one scale but the reading is taken from the other scale. Correct reading = 180° – reading (e.g. 35° should be 145°).
- **Centre not on the vertex:** place the centre point of the protractor exactly on the vertex.
- **Base arm not on the 0 line:** turn the protractor so one arm passes through 0°.
- **Protractor upside down / reflex angle:** for a reflex angle measure the smaller angle and subtract from 360°.

Figure it Out · Where are the angles?

1 (a) **At 1 o'clock, the angle between the hands of a clock is 30°. Why?**

✔ ANSWER

A full turn of 360° is divided into 12 equal parts by the 12 numbers.

Each part = 360° ÷ 12 = **30°**. At 1 o'clock the hands are one part apart.

1 (b) **What will be the angle at 2 o'clock? 4 o'clock? 6 o'clock?**

✔ ANSWER

2 o'clock: 2 × 30° = **60°**

4 o'clock: 4 × 30° = **120°**

6 o'clock: 6 × 30° = **180°** (straight angle)

1 (c) **Explore other angles made by the hands of a clock.**

💡 **SAMPLE ANSWER**

3 o'clock: 3 × 30° = 90° (right angle)

5 o'clock: 150°; 9 o'clock: 90° (or the reflex angle 270°)

12 o'clock: 0° (hands together)

2 **Can the amount a door is opened be expressed using an angle? What are the vertex and the arms?**

✔ **ANSWER**

Yes. Vertex: the hinge line where the door meets the frame.

Arms: the closed position of the door (the wall/frame line) and the door in its open position.

3 **Vidya on the swing: where is the angle? Can you see it?**

💡 **SAMPLE ANSWER**

Vertex: the point where the swing's ropes are tied at the top.

Arms: the rope when hanging straight down (rest position) and the rope at the starting position.

The angle between them is the starting angle of the swing.

4 **A toy with slanting slabs: can angles describe the slopes? What are the arms? Which arm is visible and which is not?**

✔ ANSWER

Yes. The steeper the slab, the bigger the angle.

Arms: the slab (slanting edge) and an imaginary horizontal line.

The slab is visible; the horizontal line is **not** visible.

5 **An insect and its rotated version: can angles describe the amount of rotation? What are the arms and vertex?**

✔ ANSWER

Yes. The angle shows how much the insect has turned.

Arms: the horizontal line touching the insect and the line along the rotated insect.

Vertex: the point about which it turns (where the two lines meet).

Figure it Out · 2.10 Drawing Angles

1 **In Fig. 2.23, list all the angles possible. Guess their measures, then measure with a**

protractor and record in a table.

💡 SAMPLE ANSWER

Angles: ∠CAP, ∠ACD, ∠APL, ∠DLP, ∠RPL, ∠SLP, ∠PRS, ∠LSR, ∠BRS, ∠CLP ... (find more)

Make a table: Angle | Guess | Measured, and see how close your guesses are. (check the picture in your book)

2 **Use a protractor to draw angles of: a. 110° b. 40° c. 75° d. 112° e. 134°**

✔ ANSWER

For each angle:

- Draw a base arm, e.g. IN.
- Place the protractor centre on I with IN on the 0 line.
- Using the scale that starts at 0 on IN, mark a point T at the required number (110, 40, 75, 112 or 134).
- Join I and T with a ruler. ∠TIN is the required angle.

3 **Draw an angle with the same degree measure as ∠HIJ given. Write the steps you followed.**

💡 SAMPLE ANSWER

Step 1: Measure ∠HIJ with a protractor (centre on I, IJ on 0).

Step 2: Draw a base ray, say OB.

Step 3: Put the protractor centre on O, OB on 0, and mark A at the same measure.

Step 4: Join OA. ∠AOB = ∠HIJ.

🧮 Figure it Out · 2.11 Types of Angles and their Measures

1 **In the grids, join A to other grid points to get: a. an acute angle b. an obtuse angle c. a reflex angle. Mark the angles with curves.**

💡 SAMPLE ANSWER

a. Join A to a point so the new line makes an angle smaller than an 'L' with AB.

b. Join A to a point so the angle is wider than an 'L' but less than a straight line.

c. Any acute or obtuse angle drawn has a **reflex angle** on its outside; mark the outer curve. (check the picture in your book)

2 **Measure each angle and classify: a. ∠PTR b. ∠PTQ c. ∠PTW d. ∠WTP**

💡 SAMPLE ANSWER

a. ∠PTR = 30° — **acute**

b. ∠PTQ = 60° — **acute**

c. ∠PTW = 102° — **obtuse**

d. ∠WTP (outer) = 360° − 102° = 258° — **reflex** (check the picture in your book)

⭐ Let's Explore · 2.11

? **∠TER = 80°. What is the measure of ∠BET? What is the measure of ∠SET? (∠SER = 90°)**

✔ ANSWER

∠REB is a straight angle = 180°.

∠BET = 180° − 80° = **100°**

∠SET = ∠SER − ∠TER = 90° − 80° = **10°**

🧮 Figure it Out · 2.11 (page 53)

1 **Draw angles: a. 140° b. 82° c. 195° d. 70° e. 35°**

✔ ANSWER

a, b, d, e: draw the base arm, put the protractor centre on the vertex with the arm on 0, mark the number, join.

c. 195° is reflex: 360° − 195° = 165°. Draw a 165° angle; the angle on the **outside** is 195°. Mark the outer curve.

(Or: draw a straight angle 180° and turn 15° more beyond it.)

2 **Estimate each angle, then measure it with a protractor. Classify as acute, right, obtuse or reflex.**

💡 SAMPLE ANSWER

First estimate, then measure.

Less than 90° → acute; 90° → right; between 90° and 180° → obtuse; between 180° and 360° → reflex. (check the picture in your book)

3 **Make any figure with three acute angles, one right angle and two obtuse angles.**

💡 **SAMPLE ANSWER**

Draw a hexagon-like figure: start with a right angle corner E, then corners D and F wider than 90° (obtuse), and corners A, B, C narrower than 90° (acute).

Check each corner with a protractor.

4 **Draw the letter 'M' such that the angles on the sides are 40° each and the angle in the middle is 60°.**

💡 **SAMPLE ANSWER**

Draw the two upright lines of M.

At the top of each upright, draw the slanting stroke making 40° with the upright.

Where the slanting strokes meet in the middle, they make 60°.

Check: the middle 'V' has 60° at the bottom.

5 **Draw the letter 'Y' such that the three angles formed are 150°, 60° and 150°.**

✔ **ANSWER**

Draw the vertical stem of Y upwards to point O.

At O draw two arms, each making 150° with the stem.

The angle between the two arms = 360° − 150° − 150° = **60°**.

6 **The Ashoka Chakra has 24 spokes. What is the angle between two spokes next to each other? What is the largest acute angle between two spokes?**

✔ ANSWER

Angle between neighbouring spokes = 360° ÷ 24 = **15°**.

Angles between spokes are multiples of 15°: 15°, 30°, 45°, 60°, 75°, 90°...

Largest acute angle (less than 90°) = **75°**.

7 **Puzzle: I am an acute angle. Doubling, tripling and quadrupling me give acute angles. Multiplying by 5 gives an obtuse angle. What are the possibilities?**

✔ ANSWER

4 × angle < 90° → angle < 22.5°

5 × angle > 90° → angle > 18°

So the angle is more than 18° and less than 22.5°.

In whole degrees: **19°, 20°, 21° or 22°**.

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**